

Qatar-1b

R-2063

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_121128 · created 2026-08-09T01:16:28+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456259.663694 +/- 7.8e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.136 +/- 0.052
Transit depth (Rp/Rs)^2	1.84 +/- 1.41 [%]
Orbital Inclination (inc)	89.05 +/- 3.03
Airmass coefficient 1 (a1)	1.24 +/- 0.1
Airmass coefficient 2 (a2)	-0.107 +/- 0.025
Scatter in the residuals of the lightcurve fit is	8.24 %
Best Comparison Star	#3 - [303, 80]
Optimal Method	PSF photometry
Transit Duration (day)	0.0785 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.136 ± 0.052 vs 0.1463 (deviation 0.1σ)
Duration fit vs published	0.0785 d vs 0.0692 d (deviation 0.61σ)
Mid-time O-C	0.2 min
Depth SNR	1.4
Residual scatter	8.24 %

Light curve

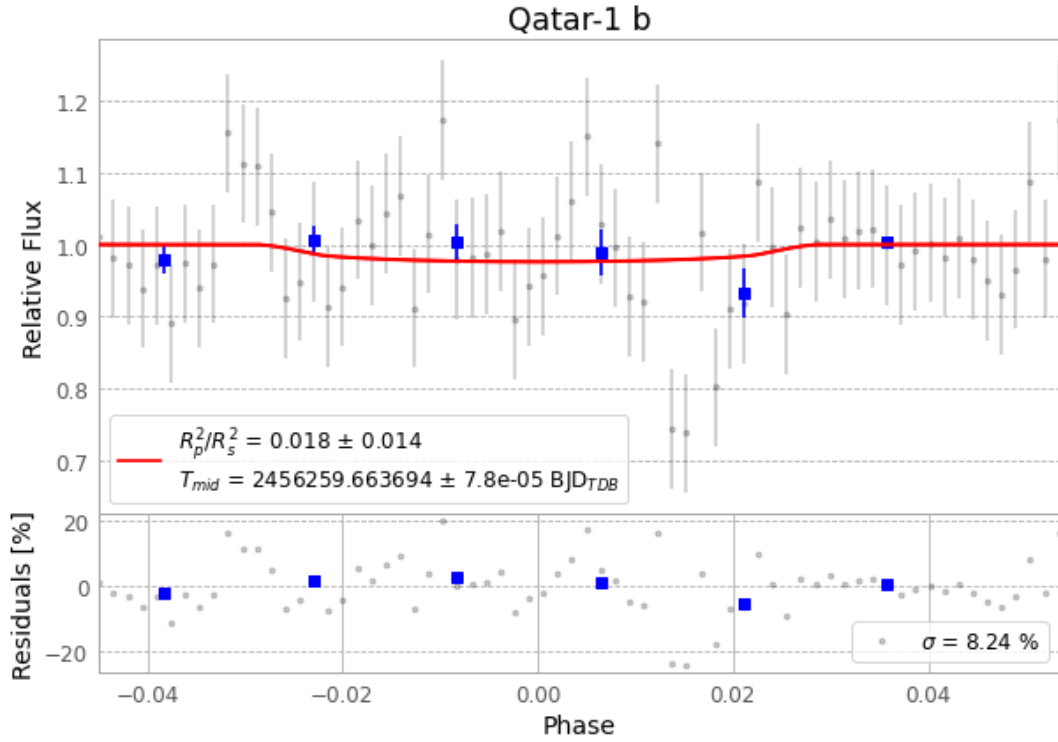


Figure 1 — FinalLightCurve_Qatar-1 b_2012-11-27.png (SHA-256 1dfaa5d1ea42237a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 271], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5c40d7c1b3bc15ea...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

70 FITS frames (46 MB), Qatar-1121128022112.FITS → Qatar-1121128054812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-121128.log (26da8465cd61...), FinalLightCurve_Qatar-1 b_2012-11-27.png (1dfaa5d1ea42...), FinalLightCurve_Qatar-1 b_2012-11-27.csv (a9a3604e0654...)

Compute resources

Wall time 171.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 01:16Z.